

Cloud Infrastructure and Security

Computing and Mathematics

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Cloud Infrastructure and Security (A13802)

Short Title: Cloud Infrastruct and Security
Faculty Science and Computing
Department Computing and Mathematics
Cluster Networks and Cloud
Author RFRISBY
Credits 10

Level: Postgraduate

Description of Module / Aims

This module will build on the student's prior knowledge of computer networks and security to provide the student with a thorough understanding of many of the latest infrastructure technologies and services. The main areas covered in this module are IP networking, network monitoring and network automation as well as the security of systems, networks and modern distributed applications.

Programmes

Indicative Content

- In depth treatment (assuming prior knowledge) of IP networking; addressing
- Routing & switching technologies
- Network monitoring & performance
- Software defined networking
- Infrastructure as Code
- Security fundamentals: cryptographic techniques for confidentiality, authentication and integrity; key management and digital certificates
- Network & system security: protocols and technologies
- Security assessment: black and white box testing; penetration testing; standards compliance
- Secure software development

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate how the latest technologies in networking and security (protocols, architectures) operate and how they can be deployed.
2. Compare and contrast major networking technologies found on the Internet today and assess their suitability for different scenarios and user requirements.
3. Design and implement solutions for application infrastructure automation.
4. Evaluate network performance using appropriate monitoring tools.
5. Assess the key aspects of system and network security in a heterogeneous environment.
6. Carry out a security assessment on a running application.
7. Implement a selection of secure software development best practices.

Learning and Teaching Methods

- Combination of lectures and computer-based practical and simulation exercises.
- The practical component will be delivered in one double lab session.
- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,7
Continuous Assessment	50%	
<i>Assignment</i>	30%	1,2,3,4,5
<i>Lab Report</i>	20%	3,4,6,7

Assessment Criteria

- <40%:** Unable to interpret and describe key concepts of the specific knowledge domains of networking, security and emerging technologies in these areas.
- 40%-59%:** Able to interpret, describe and discuss key concepts of the specific knowledge domains of networking, security and emerging technologies in these areas. Able to discover and integrate related knowledge into application architectures.
- 60%-69%:** Able to solve problems within the specific knowledge domains by experimenting with the appropriate skills and tools.
- 70%-100%:** All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	24
Lab	24	24
Independent Learning	222	222

Essential Material

- "Open Web Application Security Project (OWASP)." <https://www.owasp.org/>
- "The SANS Institute." <https://www.sans.org/>

Supplementary Material

- Kurose, J.F. and K.W. Ross. _Computer Networking: A Top-down Approach_. 6th. New York: Pearson, 2012.
- McGraw, G. _Software Security: Building Security In_. Boston: Addison-Wesley, 2006.
- Stallings, W. _Cryptography and Network Security: Principles and Practice_. 6th ed. London: Pearson, 2013.
- Stallings, W. _Foundations of Modern Networks: SDN, NFV, QoE, IoT, and Cloud_. 1st. New York: Pearson, 2016.

Requested Resources

- COMPUTER LAB: BYOD Lab